

Green-rank positivity: a master inequality and an explicit near-radial class

Working note for the “Keep the Angle” program

Purpose. Paper I reduces the rank form of Angular Preservation, for $d \leq 3$, to *Green-rank positivity* (Conjecture, “Green-rank positivity”): $\rho_G(M) := \rho_S(d_M(X, Y), -G_M(X, Y)) > 0$, where G_M is the zero-mean Green kernel of the Laplace–Beltrami operator and (X, Y) are i.i.d. from the normalized volume. Two facts are known: $\rho_G = 1$ on compact two-point homogeneous spaces (Paper I, “Uniform rank preservation”), because there G is a strictly decreasing function of distance; and $\rho_G > 0$ on a C^2 -open neighborhood of any metric with $\rho_G > 0$ (companion note, “Perturbative openness”). Both are qualitative and neither localizes the obstruction. This note gives (i) a *master inequality* bounding the rank deficit $1 - \rho_G$ by a single geometric quantity—the non-radiality of G , measured in sup norm against the best monotone radial profile—and (ii) an *explicit near-radial class* on which $\rho_G > 0$ with a quantitative bound. The two-point homogeneous result is the exact-radial ($R \equiv 0$) point of the master inequality, and the openness theorem is its small- R corollary made quantitative. Notation and imported facts follow Paper I and the companion reductions note (in particular Proposition “Distortion controls rank” and Lemma “Automatic atomlessness”); §4 states the standing assumptions and what remains open.

1 The master rank-remainder inequality

Fix M closed of dimension d , $\text{vol}(M) = 1$, with zero-mean Green kernel G , and let (X, Y) be i.i.d. from the volume. Write $W = d_M(X, Y)$ for the random pairwise distance, $D = \text{diam } M$, and let W' be an independent copy of W . All rank correlations are the population grade correlations. By Lemma “Automatic atomlessness” (companion note) the laws of $d_M(X, Y)$ and of $G(X, Y)$ are atomless (the pair law has bounded density w.r.t. $\text{vol} \otimes \text{vol}$), so no ties occur and Daniels’ inequality $|3\tau - 2\rho_S| \leq 1$ applies to every pair of the statistics below.

Definition 1. A monotone radial profile is a strictly increasing continuous function $\Psi : (0, D] \rightarrow \mathbb{R}$. Its remainder is $R(x, y) := -G(x, y) - \Psi(d_M(x, y))$, and its non-radiality is $\|R\|_\infty = \sup_{x \neq y} |R(x, y)|$. Its anti-concentration constant is the essential sup of the density of the random variable $\Psi(W)$, denoted $\phi_\Psi := \|f_{\Psi(W)}\|_\infty \in (0, \infty]$.

Theorem 2 (Master inequality). For every monotone radial profile Ψ with $\|R\|_\infty < \infty$ and $\phi_\Psi < \infty$,

$$\rho_G(M) \geq 1 - 3\mu_\Psi(2\|R\|_\infty) \geq 1 - 12\phi_\Psi\|R\|_\infty, \quad \mu_\Psi(\varepsilon) := \mathbb{P}(|\Psi(W) - \Psi(W')| \leq \varepsilon).$$

In particular $\rho_G(M) > 0$ whenever $\|R\|_\infty < 1/(12\phi_\Psi)$; and if $R \equiv 0$ (the kernel G is exactly a strictly decreasing function of distance), then $\rho_G(M) = 1$.

Proof. Let $P = (X, Y)$ and $Q = (X', Y')$ be two independent draws of the pair, with distances d_P, d_Q and Green values G_P, G_Q . Kendall's population coefficient for the pair of statistics $(d_M, -G)$ is $\tau_G = \mathbb{E}[\text{sgn}(d_P - d_Q) \text{sgn}(G_Q - G_P)] = 1 - 2p$, where $p = \mathbb{P}(\text{discordant})$ and a pair-of-pairs is discordant when $\text{sgn}(d_P - d_Q) \neq \text{sgn}((-G_P) - (-G_Q))$; atomlessness rules out ties. By Daniels' inequality $\rho_S \geq (3\tau_G - 1)/2 = 1 - 3p$, i.e. $\rho_G \geq 1 - 3p$.

Bound p . Suppose $d_P < d_Q$ (the reverse case is symmetric). Discordance means $-G_P > -G_Q$, i.e. $\Psi(d_P) + R_P > \Psi(d_Q) + R_Q$, whence $\Psi(d_Q) - \Psi(d_P) < R_P - R_Q \leq 2\|R\|_\infty$. Since Ψ is strictly increasing and $d_P < d_Q$, the left side is positive, so $|\Psi(d_P) - \Psi(d_Q)| \leq 2\|R\|_\infty$. As $\Psi(d_P) = \Psi(W)$ and $\Psi(d_Q) = \Psi(W')$ are independent copies, $p \leq \mathbb{P}(|\Psi(W) - \Psi(W')| \leq 2\|R\|_\infty) = \mu_\Psi(2\|R\|_\infty)$. This gives the first inequality.

For the second, let $A = \Psi(W)$ with density $f_A \leq \phi_\Psi$. Then for any $\varepsilon \geq 0$, $\mu_\Psi(\varepsilon) = \mathbb{E}_{A'} \mathbb{P}(A \in [A' - \varepsilon, A' + \varepsilon]) \leq \mathbb{E}_{A'} [2\varepsilon \phi_\Psi] = 2\phi_\Psi \varepsilon$, so $\mu_\Psi(2\|R\|_\infty) \leq 4\phi_\Psi \|R\|_\infty$ and $\rho_G \geq 1 - 12\phi_\Psi \|R\|_\infty$. Finally if $R \equiv 0$ then $p = 0$ (the reverse ranking is exact), so $\tau_G = 1$ and $\rho_G = 1$. $\square$

Remark 3. The two known results are the two extremes. *Two-point homogeneous:* $G(x, y) = g(d_M(x, y))$ with g strictly decreasing (Paper I), so $\Psi = -g$ has $R \equiv 0$ and Theorem 2 gives $\rho_G = 1$ —recovering the corollary with no separate argument. *Openness:* $\|R\|_\infty$ and ϕ_Ψ vary continuously under C^2 perturbation of the metric (the Green kernel and the pair-distance law do), so a metric with $\rho_G = 1$ (hence $R \equiv 0$ for its optimal Ψ) has a C^2 -neighborhood on which $\|R\|_\infty$ stays below $1/(12\phi_\Psi)$; Theorem 2 then gives $\rho_G > 0$ with an explicit floor, quantifying the qualitative openness theorem. What is genuinely new is that the inequality holds on *all* metrics, not only near the radial ones: it converts Green-rank positivity into a single sup-norm estimate on the non-radial part of G .

2 The near-diagonal singularity is radial to leading order

To turn Theorem 2 into a checkable geometric statement one needs a monotone radial profile Ψ whose remainder R is finite despite the diagonal blow-up of G . The parametrix supplies one: near the diagonal G equals the Euclidean fundamental profile plus a *bounded* regular part, and the singular coefficient is universal (independent of the point), so the diagonal contributes nothing to $\|R\|_\infty$.

Let γ_d be the Euclidean fundamental profile,

$$\gamma_3(r) = \frac{1}{4\pi r}, \quad \gamma_2(r) = -\frac{1}{2\pi} \log r,$$

(and $\gamma_d(r) = ((d-2)\omega_{d-1})^{-1} r^{2-d}$ for $d \geq 3$ in general).

Lemma 4 (Bounded regular part). *Let M be a closed smooth Riemannian d -manifold, $d \in \{2, 3\}$, $\text{vol}(M) = 1$, with injectivity radius $i_0 > 0$. Fix $r_0 < i_0$. There is a constant $H_0 = H_0(M, r_0) < \infty$ such that the zero-mean Green kernel satisfies*

$$|G(x, y) - \gamma_d(d_M(x, y))| \leq H_0 \quad \text{whenever } d_M(x, y) \leq r_0.$$

Consequently, defining $\Psi(r) = -\gamma_d(r)$ for $r \leq r_0$ and extending Ψ to $[r_0, D]$ as any strictly increasing C^1 function with $\Psi' \geq c_1 > 0$ there and $\Psi(r_0^+) = \Psi(r_0^-)$, one has, with $H_{\text{far}} := \sup_{d_M(x, y) \geq r_0} |G(x, y)| + \sup_{r \in [r_0, D]} |\Psi(r)|$,

$$\|R\|_\infty \leq \max(H_0, H_{\text{far}}) < \infty,$$

and $\phi_\Psi < \infty$.

Proof. Regular part. In geodesic normal coordinates centered at y the metric is Euclidean to second order ($g^{ij} - \delta^{ij} = O(r^2)$, drift coefficients $O(r)$). Let χ be a cutoff, $\chi \equiv 1$ on $\{d_M(\cdot, y) \leq r_0\}$, supported in the i_0 -ball. The parametrix $E_y(x) = \chi(x) \gamma_d(d_M(x, y))$ satisfies $\Delta_x E_y = \delta_y + b_y$, where $b_y = (\Delta_g - \Delta_{\text{Eucl}}) \gamma_d \cdot \chi + [\Delta, \chi] \gamma_d$ collects the Euclidean-to-Riemannian discrepancy of the leading term (the $r^{2-d}/\log r$ singularity is annihilated by the Euclidean Laplacian) and the commutator with the cutoff (smooth, supported off the diagonal). Since $(\Delta_g - \Delta_{\text{Eucl}}) \gamma_d = O(r^{2-d})$, one has $b_y \in L^p(M)$ for every $p < \frac{d}{d-2}$ (all $p < \infty$ when $d = 2$), uniformly in y ; this is where $d \leq 3$ enters. Then $h_y := G(\cdot, y) - E_y$ solves $\Delta h_y = -1 - b_y$ in the zero-mean class (from $\Delta_x G(\cdot, y) = \delta_y - 1$). Choose $p \in (\frac{d}{2}, \frac{d}{d-2})$ —a nonempty range *exactly* for $d \in \{2, 3\}$. Elliptic $W^{2,p}$ regularity and the Sobolev embedding $W^{2,p} \hookrightarrow C^0$ on the closed manifold give $\|h_y - \bar{h}_y\|_{C^0} \leq C \| -1 - b_y \|_{L^p}$, where $\bar{h}_y = \int_M h_y = - \int_M \chi \gamma_d(d_M(\cdot, y))$ is bounded uniformly in y ; hence $\|h_y\|_{C^0} \leq H_0$ uniformly. On $\{d_M(x, y) \leq r_0\}$, $\chi \equiv 1$ gives $E_y(x) = \gamma_d(d_M(x, y))$, so $|G(x, y) - \gamma_d(d_M(x, y))| = |h_y(x)| \leq H_0$.

Remainder bound. For $d_M(x, y) \leq r_0$, $R = -G + \gamma_d \circ d_M = -h_y$, bounded by H_0 . For $d_M(x, y) \geq r_0$, both G (smooth off-diagonal, hence bounded on the compact set $\{d_M \geq r_0\}$) and $\Psi \circ d_M$ are bounded by H_{far} ; hence $|R| \leq H_{\text{far}}$ there.

Finiteness of ϕ_Ψ . Ψ is a strictly increasing C^1 diffeomorphism of $(0, D]$ onto its image, so $\Psi(W)$ has density $f_W(\Psi^{-1}(s))/\Psi'(\Psi^{-1}(s))$, where f_W is the density of $W = d_M(X, Y)$ (bounded, and $f_W(r) \asymp r^{d-1}$ as $r \rightarrow 0$). Near $s \rightarrow \Psi(0^+)$ ($r \rightarrow 0$), $\Psi' = -\gamma'_d \asymp r^{1-d}$ ($d \geq 3$) or r^{-1} ($d = 2$), so the density $\asymp r^{d-1}/r^{1-d} = r^{2(d-1)} \rightarrow 0$ ($d \geq 3$), resp. $r^{d-1}/r^{-1} = r^d \rightarrow 0$ ($d = 2$); on $[r_0, D]$, $\Psi' \geq c_1 > 0$ and f_W is bounded, so the density is bounded. Hence $\phi_\Psi < \infty$. $\square$

Remark 5. The universal leading coefficient is what makes the diagonal harmless: two pairs at nearly equal small distances have Green values agreeing to leading order regardless of *where* they sit, so the singular part contributes concordantly. All the non-radiality that can hurt the ranking lives in the bounded regular part h_y (the Robin function off its radial mean) and in the mesoscopic/large-scale values H_{far} —exactly the scales at which low eigenfunctions oscillate.

3 The explicit near-radial theorem

Theorem 6 (Green-rank positivity on a near-radial class). *Let M be a closed smooth Riemannian d -manifold, $d \in \{2, 3\}$, $\text{vol}(M) = 1$, injectivity radius $\geq i_0$. Let Ψ_0 be the parametrix profile of Lemma 4, with finite non-radiality $\|R_0\|_\infty$ and finite anti-concentration constant ϕ_0 . Then*

$$\rho_G(M) \geq 1 - 12 \phi_0 \|R_0\|_\infty.$$

In particular $\rho_G(M) > 0$ —so, by the Green-kernel rank-limit theorem (Paper I), $\liminf_\Lambda \rho_S(d_M(X, Y), \|N_\Lambda(X) - N_\Lambda(Y)\|) = \rho_G(M) \geq 1 - 12 \phi_0 \|R_0\|_\infty$, the truncated commute angular ranking being uniform-in-mode-count rank-faithful with this floor—whenever

$$\phi_0 \|R_0\|_\infty < \frac{1}{12}.$$

Writing $\Delta_{\text{rad}}(M) := \inf_\Psi \| -G - \Psi \circ d_M \|_\infty$ for the best monotone-radial sup-approximation of $-G$ (the non-radiality of the Green kernel), one has $\|R_0\|_\infty \geq \Delta_{\text{rad}}(M)$, so a small non-radiality with controlled ϕ_0 suffices.

Proof. Immediate from Theorem 2 applied to the *single* profile Ψ_0 , whose non-radiality $\|R_0\|_\infty$ and anti-concentration constant ϕ_0 are both finite by Lemma 4; no optimization over profiles

is performed (in particular the bound does *not* split the product-infimum $\inf_{\Psi} \phi_{\Psi} \|R_{\Psi}\|_{\infty}$ into separate infima—an isotonic optimal profile can be flat, giving an atom in $\Psi(W)$ and $\phi_{\Psi} = \infty$). The rank-faithful consequence is the Green-kernel rank-limit theorem, which gives $\rho_S \rightarrow \rho_G$; the floor is thus approached as a $\liminf$ up to $o(1)$, not attained at finite Λ . $\square$

Remark 7 (What the class contains). On the two-point homogeneous spaces $-G$ is exactly monotone-radial ($R_0 \equiv 0$, hence $\Delta_{\text{rad}} = 0$ and $\rho_G = 1$); $\phi_0 \|R_0\|_{\infty}$ is C^2 -continuous, so the theorem covers a C^2 -neighborhood of each with a numerical floor—the effective form of the openness theorem. More broadly, any manifold with $\phi_0 \|R_0\|_{\infty} < 1/12$ is included: mild ellipsoids, low-eccentricity tori, small metric perturbations of symmetric spaces. Both $\|R_0\|_{\infty}$ and ϕ_0 are computable per metric (evaluate $-G - \Psi_0 \circ d_M$ and the density of $\Psi_0(W)$ over a volume sample), so the hypothesis is a certificate, not an article of faith. (The infimal non-radiality $\Delta_{\text{rad}} \leq \|R_0\|_{\infty}$ is a genuine geometric quantity, but it is *not* by itself sufficient: the sup-approximation optimal profile may be flat and carry $\phi = \infty$, so positivity is asserted through the fixed profile Ψ_0 , whose ϕ_0 is finite.)

Remark 8 (Sharper constants via anti-concentration of $\Psi_0(W)$). The factor $12\phi_0$ is the crude one from bounding μ_{Ψ_0} by a uniform density. When the distance law is spread— $\Psi_0(W)$ far from concentrated— $\mu_{\Psi_0}(2\|R_0\|_{\infty})$ is much smaller than $4\phi_0 \|R_0\|_{\infty}$, and the first inequality of Theorem 2 ($\rho_G \geq 1 - 3\mu_{\Psi_0}$) is the honest bound to evaluate numerically. Manifolds with a well-spread pairwise-distance spectrum tolerate proportionally larger non-radiality.

4 Standing assumptions and the frontier

Imported facts. (1) Green-function parametrix on a closed manifold: $G(x, y) = \gamma_d(d_M(x, y)) + h(x, y)$ with h bounded (indeed Hölder) up to the diagonal and γ_d the Euclidean fundamental profile—Aubin, *Nonlinear Analysis on Manifolds*, the Green-function expansion; the constant coefficient is universal. (2) Elliptic L^p regularity and Sobolev embedding on closed manifolds (Gilbarg–Trudinger; Aubin). (3) Daniels’ inequality $|3\tau - 2\rho_S| \leq 1$ (Daniels 1950), as used in Paper I’s “Distortion controls rank”. (4) Atomlessness of the pair-distance and pair-Green laws under a bounded pair density (companion note, “Automatic atomlessness”). Each is textbook or already in the program.

What is now reduced to what. Green-rank positivity is implied, with an explicit floor, by a single estimate at the fixed parametrix profile: $\phi_0 \|R_0\|_{\infty} < 1/12$ (small non-radiality $\|R_0\|_{\infty}$ with controlled anti-concentration ϕ_0). The near-diagonal singularity is discharged (Lemma 4); the estimate is a statement about the *bounded* regular and large-scale parts of G only.

Open, in order of expected difficulty. (a) A geometric upper bound on $\Delta_{\text{rad}}(M)$ in terms of $\text{Ric} \geq -(d-1)K$, diam , i_0 , and the spectral gap μ_1 : the heat-kernel route ($G = \int_0^{\infty} (p_t - 1) dt$, Li–Yau two-sided Gaussian bounds on the near part, $|p_t - 1| \leq Ce^{-\mu_1 t}$ on the far part) should yield $\phi_0 \|R_0\|_{\infty} \leq F(K, D, i_0, \mu_1)$, giving a curvature/diameter class free of the per-metric certificate. The obstacle is that the Li–Yau *constants* are not sharp, so the multiplicative near-diagonal envelope must be replaced by the sharp parametrix coefficient (Lemma 4) glued to a Li–Yau tail—routine but unpleasant. (b) Whether $\phi_0 \|R_0\|_{\infty}$ can exceed $1/12$ (so the sufficient condition lapses), and if so whether ρ_G can actually reach 0: the counterexample program on thin necks of revolution and Berger spheres, where a low mode oscillates strongly enough to make G badly non-radial. Theorem 2 says any such counterexample must have $\mu_{\Psi}(2\|R\|_{\infty}) \geq 1/3$ for *every* monotone profile, a concrete target for the search. (c) The critical dimension $d = 4$: the Green

kernel ceases to be Hilbert–Schmidt and the rank-limit theorem’s input fails; the fractional filter $\mu_k^{-s/2}$, $s > d/4$, restores it, and the first question is the analogue of Lemma 4 for the fractional kernel.